

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

5. Structured Knowledge Representation

4 hrs · asked in **38 of 41 papers** · 5 syllabus sub-topics · 319 marks, 9.4% of every mark ever set · the 11 sentence sets and the net-to-frame conversion are worked in *Ch5 – Structured Knowledge Representation – Solved Problems*.

Read this first:

- **The chapter is two objects and one comparison.** A semantic net, a frame, and net Vs frame. Learn those three and you have answered nine tenths of what this chapter has ever set.
- **The most-asked question is a drawing:** “convert the given sentences into a semantic network”, in **12 papers** across **11 sentence sets**, for 5 to 7 marks each, and **six of the eleven are the same three figures with the names changed**. Answer with a drawing, never with prose. All worked in the companion.
- **Second-most-asked, and the safest marks here:** “explain semantic net and frame with examples, list their advantages and limitations” **TOP 13/41**. One comparison table plus two small drawings.
- Three reliable short answers, each a fixed list: **issues in KR** **HOT 7/41**, **approaches to KR** **HOT 6/41**, how a KR is evaluated (four adequacies).
- **CD and scripts are only 2 papers, but both times they carry the whole 8 marks** (71 Bh **2+5**, 77 Ch **3+5**).
- Pairs the examiner reliably asks together: **issues in KR + draw the semantic network**, **approaches + semantic nets with an example**, and **compare net and frame + convert one into the other**.

5.1 Knowledge, representation, and what makes a representation “structured”

What is knowledge representation? What do you understand by structured knowledge representation? **2 2+5**

75 Bh **2**, 76 Bh **2+5** · 2 papers at 2 marks each, so no tier chip · **always the opener** to a semantic-net question, so answer it in three lines and move on

Knowledge representation

- Encoding what an agent knows about the world in a form a computer can manipulate, chosen so that **inference is easy**.
- Two things exist and must not be confused:
 - **Facts** – truths in the world. The **knowledge level**.
 - **Representations** of those facts in some formalism. The **symbol level**.
- KR is the **mapping** between the two, and it must run *both ways*: *Tommy is a dog* $\rightarrow$ `dog(Tommy)`; add $\forall x : \text{dog}(x) \rightarrow \text{hastail}(x)$, deduce `hastail(Tommy)`; map back to *Tommy has a tail*.
- **The backward map is where representations fail.** A formalism that cannot say its answer back in the user's terms is useless in an expert system.
- Three **levels** a fact sits at: **knowledge** (what is true), **logical** (the sentence encoding it), **implementation** (the data structure holding the sentence).
- The **kinds of knowledge** to name if asked: objects, events, performance (how to do a task), meta-knowledge, facts.

Structured knowledge representation

- In logic, everything known about one object is **scattered across many separate clauses**. Nothing in the formalism ties them together.
- **Structured KR concentrates all of an object's knowledge in one place**: one node of a graph, or one record of slots. The structure itself carries meaning.
- So the definition to write: **a representation in which knowledge is stored in named, linked units rather than in flat sentences**, each unit holding an object and everything known about it.
- The four structured schemes on this syllabus: **semantic nets, frames, conceptual dependency, scripts**.
- Rich and Knight call them **slot-and-filler** structures, *weak* where the slots carry no fixed meaning (nets, frames) and *strong* where they do (CD, scripts).
- **What it buys you**: inheritance, default values, and the ability to answer “what do you know about *X*” by looking in *one* place.

5.2 Approaches, models and schemes of knowledge representation

What are the different approaches / models / schemes of knowledge representation? How do you choose among them? 2 4 7 8

HOT 6/41 69 Bh 7, 80 Ba 2+5, 73 Bh 8, 79 Ash 4, 81 Bh 2, 80 Ch 2+2 · 73 Bh says “four types of KR scheme”, which is the **four-scheme list** below, not the five-method one · 80 Ch additionally asks *how you make a choice*

The four schemes (answer 73 Bh with these)

- **1. Logical** – knowledge as formal sentences, manipulated by rules of inference.
 - Eg propositional logic, first-order predicate logic (Ch 4).
 - ✓ Precise, sound, complete. × Rigid, and a single object's facts end up scattered.
- **2. Procedural** – knowledge as procedures that *do* something.
 - Eg IF-THEN production rules, an expert system's rule base.
 - ✓ Directly executable, control built in. × Knowledge and the code using it are inseparable, so nothing can be modified independently.
- **3. Network** – knowledge as a labelled graph.
 - Eg **semantic nets**, conceptual dependency, conceptual graphs.
 - ✓ Visual, inheritance is free. × No quantifiers, no negation.
- **4. Structured** – a network whose nodes are themselves *complex data structures*.
 - Eg **frames, scripts**.
 - ✓ Defaults, procedures and constraints per slot. × No inference engine of its own.
- **One sentence that ties them together:** 3 and 4 are the *structured* half of this chapter; 1 and 2 are Ch 4 and the expert system of Ch 7.

The five methods (the wording BJ Sir and PS Sir use)

- **Procedural** – facts plus the step-by-step operations on them, hard-coded.
- **Declarative** – states, facts, rules and relations declared *separately* from the algorithm that reasons over them.
 - This separation is the whole advantage: the knowledge can be changed without rewriting the reasoner, and one optimised reasoner serves many knowledge bases.
- **Relational** – facts as rows of a table, exactly as a database holds them. Simple, but carries no inference at all.
- **Hierarchical** – inheritable knowledge as a tree of classes, each level inheriting the level above.
- **Complex network graph** – the general labelled graph: semantic nets.

How to choose 80 Ch 2

- Score the candidates on the **four adequacies** of §5.3, against the domain you actually have.
- Then three practical questions:
 - Is the knowledge mostly **taxonomic**? Use a net or frames, and get inheritance for nothing.
 - Does it need **proof**, quantifiers or negation? Use logic – a net cannot express “*no bird is a mammal*”.
 - Does it need **defaults and exceptions**? Use frames; logic has no clean way to say “*birds fly, except penguins*”.
- **The examiner's expected closing line:** no single scheme wins. Real systems are **hybrid** – frames for the objects, rules for the reasoning.

5.3 Evaluating a knowledge representation, and the issues in KR

How is a knowledge representation evaluated? What are the issues in knowledge representation? 2

3 4

HOT 7/41 80 Bh 3, 80 Ch 2+2, 79 Ash 2+2, 80 Ash 3, 82 Bh 2, 76 Bh 2, 79 Bh 2 · two different lists, and **papers mix them up**: *properties* is the four adequacies, *issues* is the five design questions · 76 Bh asks the issues inside an expert-system question

Evaluating a KR: the four adequacies

- **1. Representational adequacy** – can it represent *all* the kinds of knowledge the domain needs?
- **2. Inferential adequacy** – can its structures be manipulated to *derive new* structures, ie new knowledge from old?
- **3. Inferential efficiency** – can it carry extra guidance that **focuses** the inference engine in productive directions, rather than letting it search blindly?
- **4. Acquisitional efficiency** – can new knowledge be added **easily**, ideally automatically rather than by a human?
- Two practical criteria worth adding for the last mark: it must be **easy to use**, and **easy to modify and extend**.
- **The honest closing line**: no known representation scores well on all four at once. That is exactly why hybrid systems exist, and it is what 80 Ch's "*is it possible to evaluate a KR*" is fishing for – yes, on these four axes, but only relative to a domain.

The five issues in knowledge representation

- **1. Important attributes** – are there attributes so basic that they occur in almost every domain, and so deserve special handling?
 - **Yes: isa and instance**. Both carry **property inheritance**, and every scheme in this chapter is built on them. This is the expected answer, not a shrug.
- **2. Relationships among attributes** – four to name:
 - **inverses** (*owns / owned-by*), **existence in an isa hierarchy** (*height* generalises *batting-height*), **techniques for reasoning about values** (types, ranges, rules), **single-valued attributes** (a person has one date of birth).
- **3. Level of representation, ie granularity** – how fine should the primitives be?
 - Coarse: few facts, but similar sentences look different. Fine: identical meanings collapse together, but the store explodes. CD (§5.7) is the extreme fine-grained answer.
- **4. Representing sets of objects** – by **listing members**, or by **stating the defining property**? Only the second can say "*there are lots of sheep*", which is a property of the set and of no member.
- **5. Finding the right structure** – given a huge store, how do you **select** the relevant structure and **revise** it when it turns out wrong? Indexing, matching and frame selection all live here.
- Two more the decks add: **interpretation is informal** in a net, and **complex relationships are hard** to draw with binary links.

5.4 Semantic nets

What is a semantic net? Explain with an example. When do we use one? 2 5 6 7 8

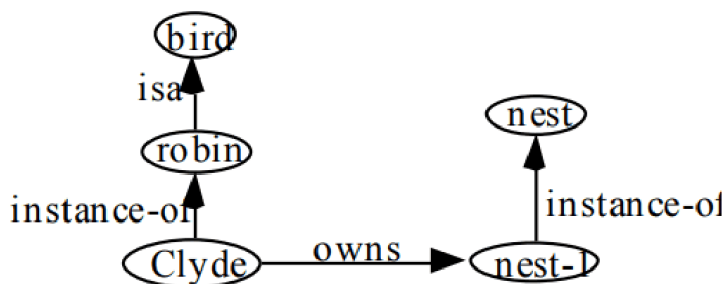
TOP 9/41 70 Bh 8, 72 Ma 7, 82 Ba 2, 75 Bh 6, 71 Bh 7, 69 Bh 7, 80 Ba 5, 76 Bh 5, 80 Bh 5 · a further 12 papers ask you to draw one, all worked in the companion · 72 Ma asks for a net about the AI course itself

What it is

- A **directed labelled graph** that stores knowledge as **nodes** and **arcs**.
 - **Nodes** = objects, concepts, situations or events.
 - **Arcs** = the relationship between two nodes, and the arc is labelled with that relationship.
- A **declarative graphic representation**: it states what is true and supports automated reasoning over it, but contains no procedure.
- Proposed as **an alternative to predicate logic**. Same facts, but all of one object's facts meet at one node instead of being scattered over clauses.
- **Oldest KR in the subject**: the Tree of Porphyry, 3rd century AD. Worth a line in an 8-mark answer.

The two links every paper wants named

- **isa / a-kind-of** – **class to class**. *A cat is a mammal*. Set inclusion: $Cats \subset Mammals$.
- **instance / member** – **individual to class**. *Tom is a cat*. Set membership: $Tom \in Cats$.
- **Getting these two the wrong way round is the commonest lost mark** in the drawing questions. *Sakti Gauchan is an instance of Person; Person is isa Mammal*.
- Everything else is an ordinary relation arc: **has-part**, **owns**, **likes**, **team**, **uniform-color**.



isa runs class to class, instance-of runs individual to class, and owns is an ordinary relation. Source: BA Sir CH-05.

The canonical net, and its FOPL

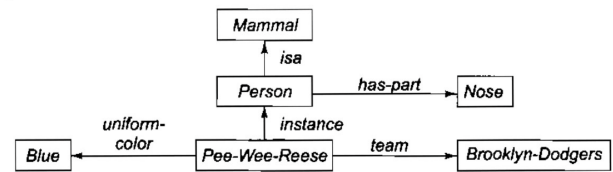


Fig. A Semantic Network

isa(Person, Mammal)
instance(Pee-Wee-Reese, Person)
team (Pee-Wee-Reese, Brooklyn-Dodgers)
uniform-color(Pee-Wee-Reese, Blue)

Rich and Knight's figure. Every "convert these sentences" question in the papers is this net with the names changed: a class chain at the top, an individual hung off it by instance, and its own properties fanning out. Source: PS Sir ch5.

Why a net can do things logic cannot

- **Inheritance is free**. A property put on *Person* is automatically available to *Pee-Wee-Reese* by following *instance* then *isa* upward. No inference rule is needed; the arcs are the rule.
- **Intersection search**. Spread activation outward from two nodes and see where the two wavefronts meet. That is how a net answers "what connects the Brooklyn Dodgers and blue?" – a question logic can only answer by proving a conjecture you already had to guess.
- **This is the advantage to quote**, because it exploits the entity-based organisation that slot-and-filler structures have and logic does not.

Advantages and limitations of a semantic net; how do you represent a three-place relation? 2 4

part of the *advantages and limitations* question in 73 Ma, 76 Ba, 78 Ch, 78 Ba, 81 Bh · chip sits on \$5.6, where the comparison is

Advantages

- **Natural and transparent:** close to how people link ideas, so it is readable by a non-programmer.
- **Easy to visualise,** and easy to check by eye.
- **Inheritance** of properties comes for free down *isa* chains.
- **Space-efficient:** each object appears *once*, however many facts mention it.
- Related facts sit next to each other, so **relevant knowledge is found by following links** instead of by searching the whole store.

Limitations

- **Cannot express quantification, negation or disjunction.** There is no way to draw “every mammal has some parent”, “no bird is a mammal” or “Tom is black or white”. This is the limitation to write first.
- **No standard meaning for link names.** *has-part* means whatever the person who drew it meant, so two nets are not comparable.
- **No inference mechanism of its own:** it stores, it does not deduce.
- **Slow at query time** – answering may need a traversal of the whole graph.
- **Does not scale:** a net that really modelled human memory cannot be built by hand.
- Interpretation is **informal**, so the net depends entirely on its creator.

Three-place relations: use an event node

- Arcs are **binary**, so “John gave the book to Mary” cannot be one arc: it has three arguments.
- **The fix:** create a **new node for the event itself**, then hang one binary arc on it per argument.

John gave the book to Mary.

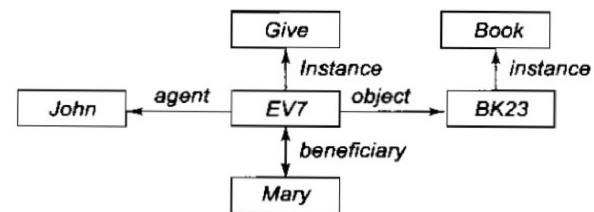


Fig. A Semantic Net Representing a Sentence

EV7 is the giving-event node. *agent* → John, *object* → the book, *beneficiary* → Mary, and *instance* says EV7 is a Give. Source: PS Sir ch5.

- Same trick for a baseball game with two teams and a score: *isa*(G23, Game), *home_team*(G23, Dodgers), *visiting_team*(G23, Cubs), *score*(G23, 5-3).
- **Say why it works:** it converts an *n*-place predicate into *n* binary ones by *reifying* the relation as an object.

What do the arcs of a semantic net actually mean? 2

not asked on its own · included because it is the **one line that separates a 5-mark answer from a 7-mark one** on every “explain semantic nets” question

Link Type	Semantics	Example
$A \xrightarrow{\text{Subset}} B$	$A \subset B$	Cats $\subset$ Mammals
$A \xrightarrow{\text{Member}} B$	$A \in B$	Felix $\in$ Cats
$A \xrightarrow{R} B$	$R(A,B)$	Felix $\xrightarrow{\text{Friend}}$ Opus
$A \xrightarrow{\boxed{R}} B$	$\forall x \ x \in A \Rightarrow R(x,B)$	Birds $\xrightarrow{\boxed{\text{Legs}}}$ 2
$A \xrightarrow{\boxed{R}} B$	$\forall x \ x \in A \Rightarrow \exists y \ y \in B \wedge R(x,y)$	Bird $\xrightarrow{\boxed{\text{Parent}}}$ Birds

- Subset arc = **set inclusion**, $A \subset B$. Eg *Cats* $\subset$ *Mammals*.
- Member arc = **set membership**, $A \in B$. Eg *Felix* $\in$ *Cats*.
- A named arc between two *individuals* = the ground fact $R(A, B)$. Eg *Friend*(Felix, Opus).
- A named arc drawn out of a *class* box means $\forall x (x \in A \Rightarrow R(x, B))$: **every** member has that value. Eg *Legs* out of *Birds* → 2.
- Same arc with an existential reading: $\forall x (x \in A \Rightarrow \exists y (y \in B \wedge R(x, y)))$. Eg *Parent* out of *Bird* → *Birds*: every bird has *some* bird parent.
- **Why it matters:** the last two are the only quantification a net can express, and only because the reader agrees to read the picture that way. Nothing in the drawing says which of the two is meant. That ambiguity is the “lack of formality” limitation.
- Source: BJ Sir [AI_Chapter5_new](#).

5.5 Frames

What is a frame? Explain with a suitable example. What is its significance in AI? **1 5 7 8**

HOT 5/41 69 Po **7**, 74 Bh **7**, 71 Bh **8**, 82 Ka **1**, 80 Ash **5** · 82 Ka wants **one sentence** for 1 mark · 80 Ash asks for the *importance* of frames, so end on the advantages

Definition

- A frame is a **collection of attributes and their values that describes one entity**, together with that entity's relationship to other frames.
 - **Slot** = the attribute name. **Filler** = its value.
 - So a frame is a **record structure with meaning attached**.
- A **single frame is almost useless**. The power is in a **frame system**: many frames, connected because **the value of one frame's slot is another frame**.
- **The line that earns the mark**: a semantic net is a **two-dimensional** representation, nodes and arcs; **frames add a third dimension by letting a node have internal structure**.
- A frame is designed to represent a **stereotypical** object or situation: what is normally true, with defaults for what has not been observed.
- Frames are **static**. A script (\$5.7) is the same idea made *sequential*.

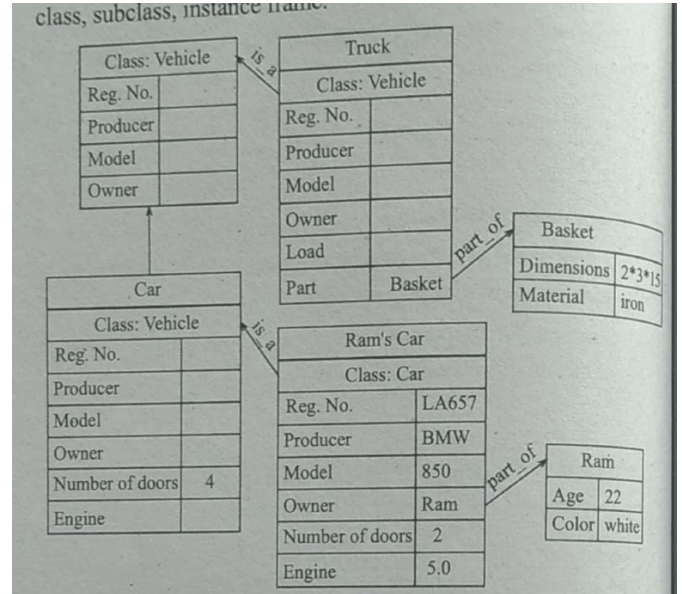
The four parts of a frame

- **1. Frame identification name** – what entity this frame is about.
- **2. Relationship to other frames** – *isa*, *instance*, *part-of*, *superclass*.
- **3. Slot and filler** – the attribute–value pairs themselves.
- **4. Default information** – values taken as true *until contradicted*. **Logic has no clean way to do this**; frames do, and it is the strongest single argument for them.

Types of frame

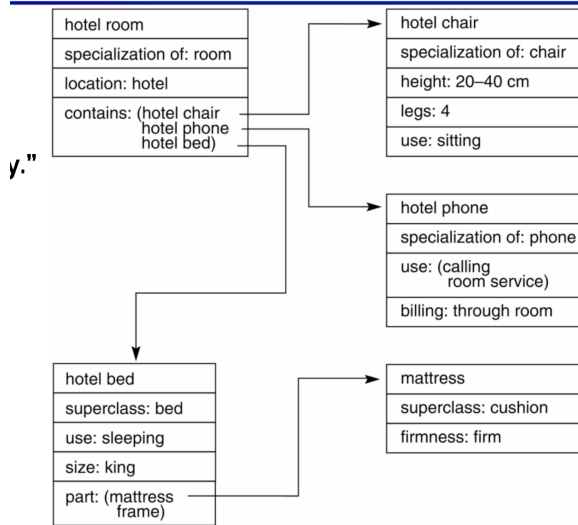
- **Class frame** – the generic frame others inherit from. Eg *Vehicle*.
- **Subclass frame** – inherits a class frame and adds slots. Eg *Car*.
- **Instance frame** – one individual. Eg *Ram's Car*.
- An instance or subclass frame **inherits every slot above it** and may add its own. Generic frames sit at the top of the hierarchy.

Class, subclass and instance in one picture



Vehicle (class) → *Truck* and *Car* (subclass, *is-a*) → *Ram's Car* (instance), with *part_of* links out to *Basket* and *Ram*. Note the instance frame fills in the slots it inherited empty. Source: *Insights* book p136.

Part of a frame description of a hotel room



Part of a frame description of a hotel room. Source: BA Sir CH-05.

What can a slot hold, and what are demons? 2 4

inside the frame question in 71 Bh 8 and 69 Po 7 · a slot holds more than a value, and saying so is the difference between a 5-mark frame answer and an 8-mark one

Example Frame: STUDENT

Slot	Value (Current Filler)	Default (Default Filler)	Cardinality (Min..Max)	Type (Type Restriction)	Procedure (Attached Procedure)	Salience (Importance)	Constraint (Rule / Axiom)
Name	Ram	–	1 .. 1	String	–	High	Cannot be empty
Age	20	18	1 .. 1	Integer	If-added: Verify age	Medium	Age ≥ 16
Department	Computer Engineering	Computer Engineering	1 .. 1	Department	–	High	Must be a valid department
Phone Number	9841234567	–	1 .. 3	String	If-added: Validate format	Low	Must contain 10 digits
CGPA	3.75	0.0	1 .. 1	Float	If-needed: Calculate from marks	High	0 ≤ CGPA ≤ 4.0

• Value: current value stored in the slot.
 • Default: value used if no value is provided.
 • Cardinality: minimum and maximum number of fillers.
 • Type: type restriction on the fillers.
 • Procedure: attached procedures (if-needed, if-added, if-removed).
 • Salience: measure of importance.
 • Constraint: rule that the value must satisfy.

One STUDENT frame with every facet filled in. Source: PS Sir ch5.

Slots	Fillers
name	property
specialization_of	a_kind_of object
types	(car, boat, house) if-added: Procedure ADD_PROPERTY
owner	default: government if-needed: Procedure FIND_OWNER
location	(home, work, mobile)
status	(missing, poor, good)
under_warranty	(yes, no)

A generic property frame carrying two demons:

if-added: ADD_PROPERTY on types,
if-needed: FIND_OWNER on owner, which also shows a default of government. Source: PS Sir ch5.

A frame system: four frames pointing at each other

- hotel room carries three ordinary slots (*specialization of*, *location*, *contains*), and the third one's filler is **three pointers, not three strings**. That is what makes this a frame system.
- Each frame it points at repeats the pattern. *hotel chair* has *specialization of: chair*, so it inherits everything a chair has and **overrides only height, legs and use**.
- *hotel bed* adds a *part* slot filled by the *mattress* frame, which is **part-of**, not **is-a**. **Confusing those two is the commonest error in a frame answer**: *is-a* inherits, *part-of* does not.
- **The line to write under your own drawing**: the diagram is a graph whose nodes happen to be records. That is structured knowledge representation in one picture, and it is why 80 Bh can ask for a net and a frame in the same breath.

Facets: what a slot carries besides its value

- **Value** – the current filler.
- **Default** – the filler used when nothing has been observed.
- **Cardinality** – minimum and maximum number of fillers allowed.
- **Type** – a restriction on what a filler may be, usually another frame.
- **Procedures** – attached code, see demons below.
- **Salience** – how important this slot is to the frame.
- **Constraints** – axioms the value must satisfy, eg $0 \leq CGPA \leq 4.0$.

Demons (attached procedures, Insights §5.4.3)

- A demon is a **procedure stored in a slot and fired automatically when the slot is touched**. The frame does not just hold data, it reacts.
- **IF-NEEDED** – fires when a value is required and none is stored. *Compute it now*.
- **IF-ADDED** – fires when a value is put into the slot. *Propagate the consequences*.
- **IF-REMOVED** – fires when a value is deleted.
- **IF-CHANGED** – fires when a stored value is replaced.
- **Why this matters for the exam**: demons are what make a frame system *active*, and they are the honest answer to “frames have no inference mechanism” – they have no *general* one, but each slot can carry its own.
- **Frames Vs objects**: slots are data members, demons are methods, *isa* is inheritance and *part-of* is containment. **A frame system is object-oriented programming, invented for AI a decade earlier**. One sentence, and it is often worth a mark.

Advantages and limitations of frames 2 4

part of the *advantages and limitations* question, chip on \$5.6 · **write these as a pair with the semantic net list**, because every paper that asks for one asks for the other

Advantages

- **Groups related knowledge together**, so programming and retrieval are easier: everything about an object is in one record.
- **Defaults**, and therefore the ability to reason with incomplete information and to detect missing values.
- **Easy to extend**: adding a slot, a property or a relation costs one line.
- **Inheritance** down the class hierarchy, exactly as in a semantic net.
- **Procedural attachment**: a slot can compute or validate itself.
- **Flexible and widely used** – vision and natural language processing both run on frames.
- Good at **causal knowledge**, because the information is organised by cause and effect.

Limitations

- **No inference mechanism of its own**. Frames store and inherit; they do not deduce. Any reasoning has to be bolted on.
- **Modification is dangerous**: change one frame and every frame that inherits from it is affected.
- **Redundancy** – the same fact easily ends up in several frames.
- **No standard structure**: the frame for a room and the frame for a person look nothing alike, so nothing generic can be written over them.
- **Defaults can be wrong**, and nothing in the formalism flags it when they are.
- **The examiner's expected trade-off line**: frames win on organisation and defaults, logic wins on proof, nets win on visibility.

5.6 Semantic nets Vs frames, and converting one into the other

Explain semantic nets and frames with examples. Compare them, and list their advantages and limitations 2 4+4 6+2 7

TOP 13/41 72 Ash 7, 72 Ma 7, 73 Ma 4+4, 74 Ma 2+5, 74 Bh 7, 74 Bh 4+3, 75 Ba 2+6, 76 Ba 6+2, 78 Ba 6+2, 78 Ch 4+4, 79 Jth 5, 81 Ba 3+5, 81 Bh 2 · **the second-most-asked question in the chapter** · 72 Ash and 78 Ba ask the **"why are they important in AI"** follow-up, so keep two lines for it

	Semantic net	Frame
Shape	labelled directed graph	collection of records , slots and fillers
Dimension	two : node and arc	three : node, and structure <i>inside</i> the node
Unit of knowledge	one arc = one fact	one frame = one whole object
Node holds	a name only	slots, values, defaults, types, constraints, procedures
Relations	drawn as arcs	written as slot values that are other frames
Inheritance	✓ down <i>isa</i> / <i>instance</i> arcs	✓ down class → subclass → instance
Defaults	× cannot express	✓ first-class, with <i>if-needed</i> to compute one
Procedures	× none	✓ demons on any slot
Best at	seeing the relationships at a glance	holding everything about one object in one place
Weakest at	complex or structured facts about a single node	showing the shape of the knowledge as a whole
Reasoning	intersection search, link following	inheritance and slot lookup only
Both fail at	quantification, negation, disjunction, and having no inference engine of their own	

How frames are useful in semantic nets 72 Ash 7

- The question sounds like two topics. It is one: a **frame** is what you get when you let a semantic-net node have contents.
- A net node called *Tom* can only be named. Make it a frame and it can carry *colour: ginger*, *age: default 2*, *owner: John*, plus a demon that computes age from a birth date.
- So frames **fix the net's weakest point**: they let one node hold structured, defaulted, constrained knowledge instead of forcing every detail out onto its own arc.
- And the net **fixes the frame system's weakest point**: it shows the shape of the whole knowledge base, which a pile of records does not.
- **Answer in one sentence**: frames give a semantic net depth at each node, and the net gives a frame system its map.

Why both matter in AI 78 Ba 2

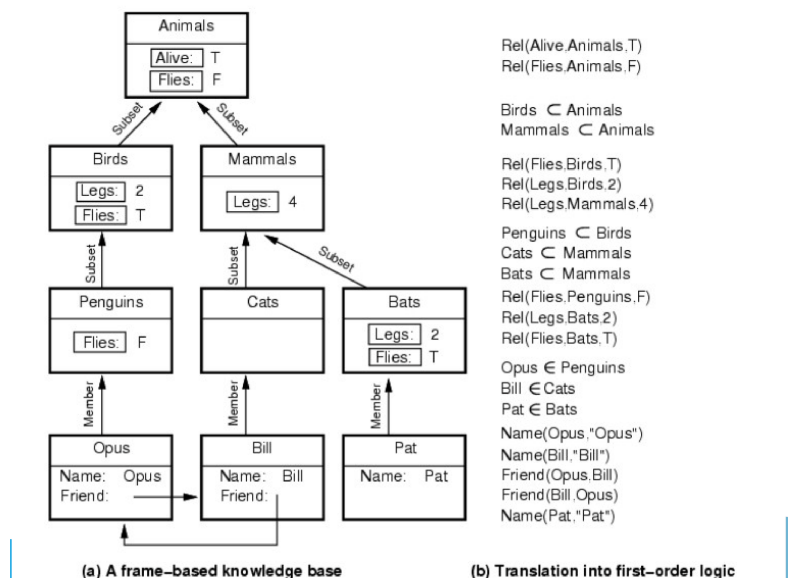
- They give **inheritance and defaults**, which is how an agent reasons with incomplete knowledge.
- They are **human-readable**, so a domain expert can check the knowledge base without knowing logic – the bottleneck in every expert system (Ch 7).
- They organise knowledge by **object** rather than by *sentence*, so retrieval is local instead of a search of the whole store.

The same knowledge three ways

- **Frame, net and FOPL are interchangeable** for taxonomic knowledge, and 75 Ba 6 asks for exactly that: *examples of both, with FOPL statements*.
- Read the figure left to right: a frame-based knowledge base, then the first-order sentences that say the same thing.
- `Subset(Birds, Animals)` is the *isa* arc. `Member(Opus, Penguins)` is the *instance* arc. `Rel(Legs, Birds, 2)` is a slot with a value.
- **Write the FOPL line beside every arc you draw** in that question. It is 2 free marks and most answers skip it.

Converting a semantic net into a frame 81 Ba 5

- **Step 1.** Every node becomes a **frame**, named after the node.
- **Step 2.** Every **outgoing arc** of that node becomes a **slot** in its frame. **The arc label is the slot name, the arc's target is the filler.**
- **Step 3.** *isa* and *instance* arcs become the frame's **class link**, not ordinary slots: *isa* → a subclass frame, *instance* → an instance frame.
- **Step 4.** Any slot whose filler is a node that has its own frame is **left as a pointer to that frame**, which is what makes the result a frame system.
- **Step 5.** Push each property to the **highest** class that has it, and let everything below inherit. A property drawn on three individuals in the net belongs on their class in the frame system.
- **Check:** no information may be lost. Count the arcs in the net; count the filled slots in the frames. They must match.
- 81 Ba is worked end to end in the companion, on this net.



(a) a frame-based knowledge base and (b) its translation into first-order logic, the same nine facts twice. Source: PS Sir ch5.

5.7 Conceptual dependency and scripts

What is conceptual dependency? Explain the common primitives used in it 2+5 3

PIN 2/41 71 Bh 2+5, 77 Ch 3+5 · only 2 papers, but **both times CD and scripts together carry the whole 8-mark question** · 77 Ch pairs it with "how is knowledge represented using scripts"

What it is

- Schank, 1973–75, built for **natural language understanding**.
- A representation of the **meaning** of a sentence, built from a **small fixed set of primitive acts** instead of from the sentence's own words.
- **Three goals**, and the papers want all three:
 - to **help draw inferences** from a sentence;
 - to be **independent of the words** used in the input;
 - so that **any two sentences identical in meaning have exactly one representation**.
- Hence *I gave the man a book, the man got a book from me* and *the book was given to the man by me* all draw the **same** diagram. That is the point of the whole theory.
- A **strong** slot-and-filler structure: unlike a frame, the slots have **fixed, agreed meanings**.

The four conceptual categories

- **ACT** – an action, always one of the primitives below.
- **PP** – *picture producer*, an object or actor.
- **AA** – *action aider*, a modifier of an ACT (how, when, where).
- **PA** – *picture aider*, a modifier of a PP (colour, size, quality).
- **Dependency rule**: PP and ACT are **governors**, PA and AA are **dependents**. A dependent cannot stand alone; a conceptualization needs at least two governors.
- Eg *Sally stroked her fat cat*: PP = Sally, cat; ACT = stroke; PA = fat.

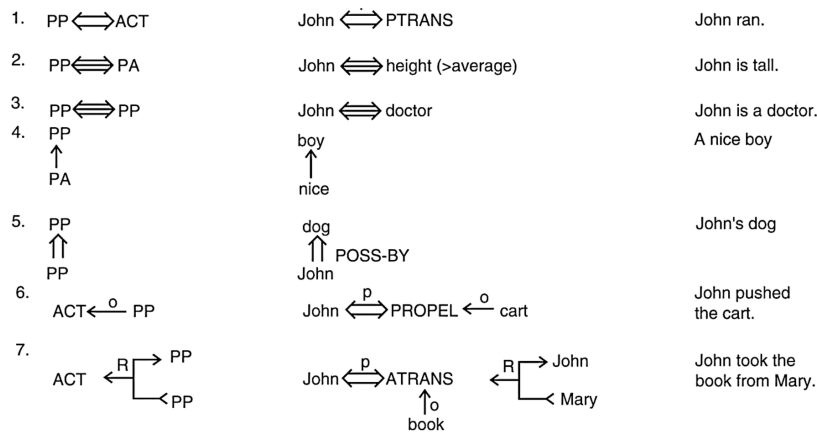
The eleven primitive acts

Primitive	Transfer or action, and a verb that maps to it
ATRANS	transfer of an abstract relationship , ie possession. <i>give, lend, borrow</i>
PTRANS	transfer of the physical location of an object. <i>go, come, went</i>
PROPEL	application of a physical force to an object. <i>push, pull</i>
MOVE	movement of a body part by its owner. <i>kick, punch</i>
GRASP	an actor grasping an object. <i>clutch, catch</i>
INGEST	an animal taking something in . <i>eat, drink</i>
EXPEL	an animal forcing something out . <i>cry, sweat, throw</i>
MTRANS	transfer of mental information . <i>tell, ask, read</i>
MBUILD	building new information out of old. <i>decide, infer, conclude</i>
SPEAK	producing a sound . <i>say, talk, play music</i>
ATTEND	focusing a sense organ on a stimulus. <i>listen, watch</i>

- **Learn the first three and MTRANS**. Every example in every paper and every scene of the restaurant script is built from those four.

Draw the conceptual dependency of a sentence 2 5

inside 71 Bh 5 · the notation cannot be typed, so **learn it as a picture**



The seven dependency forms, then each one used on a sentence. Row 6 is *John pushed the cart* (ACT $\xleftarrow{o}$ PP, PROPEL); row 7 is *John took the book from Mary* (ATRANS with a recipient pair). Source: BA Sir CH-05.

The notation

- $\longleftrightarrow$ **two-way link** between an actor and the action. Every conceptualization has exactly one.
- $\rightarrow$ **direction of dependency**: the dependent points at its governor.
- **o** = objective case, what the act is done to.
- **R** = recipient case, drawn as a **fork**: the arrow up is the receiver, the arrow down is the giver.
- **I** = instrumental case, what it was done with.
- **D** = directive case, from where to where.
- **p** = past tense, **f** future, **t** transition, **?** interrogative, **/** negative. Tense sits on the $\longleftrightarrow$.
- **Worked example** – *I gave a book to Ram*:
 - *I* $\longleftrightarrow$ ATRANS, with **p** on the link;
 - $\xleftarrow{o}$ book;
 - $\xleftarrow{R}$ forking up to *Ram* and down to *I*.

Problems with CD

- **The original sentence cannot be reconstructed** from the diagram; meaning survives, wording does not.
- **Choosing the primitive is hard**: one verb may map to several ACTs depending on context.
- **The structures get huge**. “*John bet Mike that Nepal will win the World Cup*” needs an enormous diagram for one sentence.
- Rules must be **hand-written per primitive**, so the theory does not scale to a general-purpose KR.

How is knowledge represented using scripts? Explain with an example 5

77 Ch 5 · 1 paper at 5 marks, so no chip of its own · **always asked as the second half of the CD question**, never alone · the restaurant script is the only example any deck or the textbook uses, so learn that one

What a script is

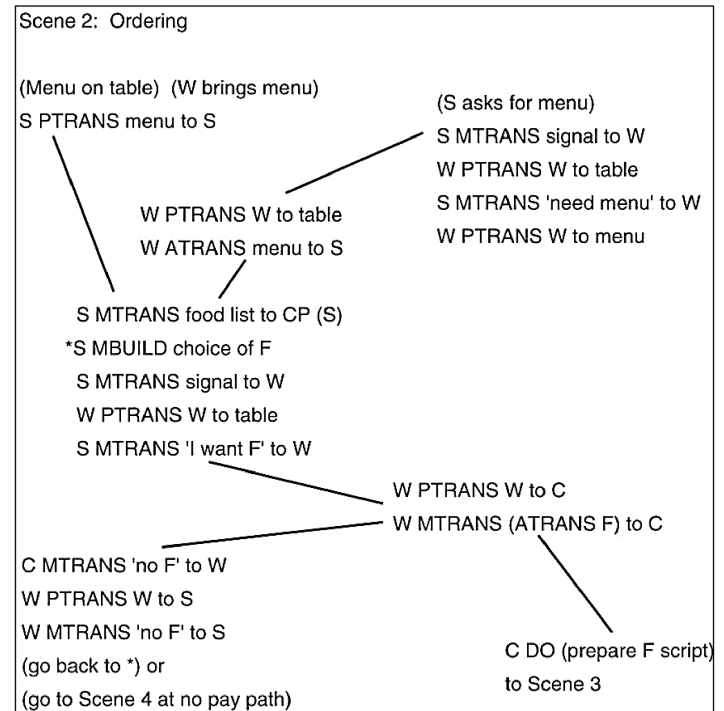
- **Schank and Abelson, 1977**, written in the CD notation.
- A **structured description of a stereotyped sequence of events in a known context**: going to a restaurant, going to the theatre, visiting a doctor.
- It **extends a frame** by making the slot values **ordered in time** and by representing *expectations* about what happens next.
- Justified by three facts about the world: events occur in known patterns, events cause one another, and entry conditions decide whether a pattern applies at all.

The six components (write all six)

- **Entry conditions** – must be true before the script can be invoked. Eg *S is hungry, S has money*.
- **Results** – true once it has run. Eg *S is not hungry, O has more money*.
- **Props** – the objects involved. Eg tables, menu, food, check, money.
- **Roles** – the participants and what each does. Eg *S = customer, W = waiter, C = cook, M = cashier, O = owner*.
- **Track** – a variation on the general pattern. Eg *coffee shop Vs fancy restaurant*: same script, different track.
- **Scenes** – the ordered sequence of events, each written in CD primitives.

The RESTAURANT script, header and scene 1

- **Script**: RESTAURANT. **Track**: coffee shop.
- **Props**: tables, menu, F = food, check, money.
- **Roles**: S = customer, W = waiter, C = cook, M = cashier, O = owner.
- **Entry conditions**: S is hungry, S has money.
- **Results**: S has less money, O has more money, S is not hungry, S is pleased (optional).
- **Scene 1, entering**: S PTRANS S into restaurant · S ATTEND eyes to tables · S MBUILD where to sit · S PTRANS S to table · S MOVE S to sitting position.



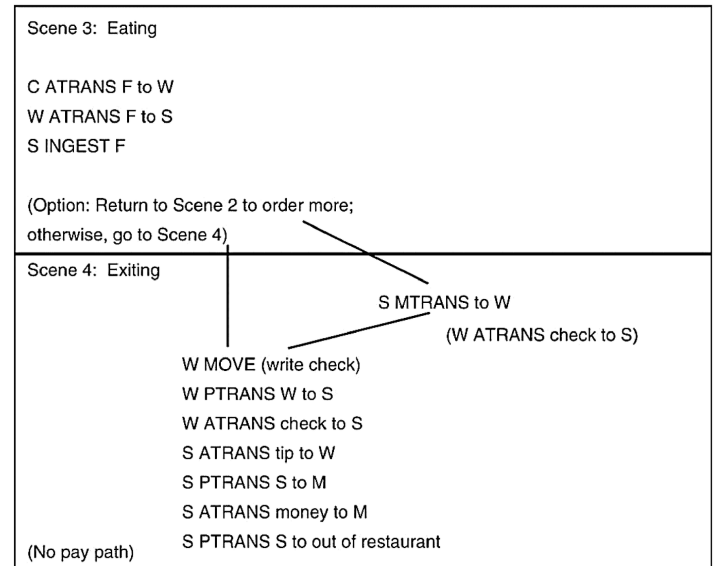
Scene 2, ordering, with both tracks: menu already on the table, or the waiter brings it. Source: BA Sir CH-05.

Invocation, and why it is the hard part

- A script is opened when its topic is **significant** in the input; if merely mentioned, only a pointer is kept.
- “John enjoyed the play in the theatre” → invoke, and every implicit question is answered: did he go, did he buy a ticket, did he have money.
- **The failure case to quote:** “John went to the theatre to pick up his daughter”. Same words, low significance, and invoking the script produces confident wrong answers.
- There is no reliable rule for judging significance, only heuristics. That is the honest limitation to close on.

Advantages and disadvantages

- ✓ **Predicts implicit events:** it fills in what the text never said, which is exactly what language understanding needs.
- ✓ Builds **one coherent interpretation** out of scattered observations.
- × **Far more specific and less general than a frame.** One script covers one situation.
- × Not suitable for knowledge that is not event-sequenced.
- × Wrong invocation gives confidently wrong answers.
- The cure for the inflexibility: **MOPs**, memory organization packets – smaller reusable modules recombined to fit the situation.



Scene 3 eating and scene 4 exiting, including the *no pay* path. Source: BA Sir CH-05.

Source notes. *Insights* §5.4.2 (book p135) converts the Tom-the-cat semantic net into frames, but its figure prints like John and type Ginger where the net says *is-owned-by* and *is-coloured*, and gives the John frame an Age 22 / Color white pair carried over from the car example on the facing page. The net-to-frame recipe in §5.6 above follows the book's *method* and Rich and Knight's *figures*, and the companion works the conversion on a net that is drawn correctly. · The syllabus files this chapter as five sub-topics but the papers only ever set four: representations and mappings is never asked on its own, always as the two-mark opener to *what is knowledge representation*.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = CT 653 (BCT, III-II, 26 papers) · monospace = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.